

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The function f is defined by $f(x) = 4x + k(x - 1)$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

Correct Answer: 67

Rationale

The correct answer is **67**. It's given that $f(5) = 32$. Therefore, for the given function f , when $x = 5$, $f(x) = 32$. Substituting **5** for x and **32** for $f(x)$ in the given function $f(x) = 4x + k(x - 1)$ yields $32 = 4(5) + k(5 - 1)$, or $32 = 20 + 4k$. Subtracting **20** from each side of this equation yields $12 = 4k$. Dividing each side of this equation by **4** yields $k = 3$. Substituting **3** for k in the given function $f(x) = 4x + k(x - 1)$ yields $f(x) = 4x + 3(x - 1)$, which is equivalent to $f(x) = 4x + 3x - 3$, or $f(x) = 7x - 3$. Substituting **10** for x into this equation yields $f(10) = 7(10) - 3$, or $f(10) = 67$. Therefore, the value of $f(10)$ is **67**.